MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU- UGP (HONOURS) Regular examination, OCTOBER 2025
THIRD SEMESTER
Discipline Specific Elective (DSE) MG3DSEBOTA08- SOIL SCIENCE
(2024 Admission onwards)
Time: 2 Hours Maximum marks: 70
Q. Paper Code: 25900267

Answer key

PART A
Answer any 10 questions (Each question carries 2 marks)

1. How does soil texture influence pore space?
   Soil texture — the relative proportion of sand, silt, and clay particles — has a major influence on **pore space** (the volume of soil not occupied by solid particles). This is because particle size and arrangement determine how tightly particles pack together and how large or small the pores between them.
2. Name one leguminous crop used as green manure.
   Sunhemp (Crotalaria juncea), Cowpea (Vigna unguiculata), Clover (Trifolium spp.). Any other.
3. Which of the following is considered a macronutrient for plant?
   Nitrogen, phosphorous, potassium or any other.
4. Which form of soil water drains quickly after rain fall?
   Gravitational water
5. Give one example of a nitrogenous fertilizer.
   Ammonium sulfate ($(NH_4)_2SO_4$), Ammonium nitrate (NH_4NO_3), Calcium ammonium nitrate (CAN)
6. Define soil fertility.
   Soil fertility is the capacity of soil to provide essential plant nutrients in available form and in sufficient quantity to support healthy plant growth.
7. Name one characteristic of good soil tilth.
   Good soil structure. Adequate moisture retention, Proper aeration, Ease of cultivation or any other.
8. Short note on Jenny's concept of soil formation.
   Soil is formed as a function of several independent factors, which together determine its properties.
9. What are primary minerals? Give one example.
   Primary minerals are minerals that have not been altered chemically since their formation; they are derived directly from the parent rock during the solidification of magma.
   Examples: Quartz, Feldspar, Mica, Hornblende. Any other.
10. What is soil science?
    Soil science is the study of soil as a natural resource, including its formation, classification, physical, chemical, and biological properties, and its relationship to plant growth and the environment.
11. What is the molecular structure of water molecule?
    H2O
12. Explain soil texture.
    Soil texture refers to the proportion of different-sized mineral particles—sand, silt, and clay—in a soil. It is a key physical property of soil that influences water retention, aeration, nutrient availability, and root growth.

PART B

Answer any 8 questions (Each question carries 5 marks)

13. Differentiate soil profile and soil horizon.

    Soil Profile: A soil profile is a vertical section of the soil from the surface down to the unaltered parent material, showing all its distinct layers. It provides a snapshot of soil formation, composition, and structure.

    Soil horizon: A soil horizon is a distinct layer of soil within a soil profile that has different physical, chemical, and biological characteristics from the layers above and below it. Each horizon forms due to soil-forming processes like organic matter accumulation, leaching, and mineral weathering.

14. How does water behave in soil?

    Water in soil exists in different forms and moves differently depending on soil texture, structure, and gravity. Its behavior is crucial for plant growth, nutrient movement, and soil processes.

15. What is soil air and how does its composition differ from atmospheric air?

    Soil air is the gaseous phase present in the pore spaces of soil, which fills the spaces not occupied by water or soil particles. It is essential for root respiration, microbial activity, and overall soil health. Soil air has lower concentration of oxygen and higher concentration of CO2 compared to atmospheric air due to root respiration and microbial activity.

16. What does the term “time” refer to in soil formation and why is it classified as a passive factor?

    In soil science, time refers to the duration over which soil-forming processes act on parent material to develop distinct soil properties and horizons. It is essentially the age of the soil and the length of exposure to factors like climate, organisms, and topography. Time considered as a passive factor because time does not actively cause soil formation; instead, it allows other factors (climate, organisms, relief, parent material) to work.

17. What is the definition of weathering in Geology?

    Weathering is the natural process by which rocks and minerals at or near the Earth’s surface are broken down into smaller particles through physical, chemical, or biological processes.

18. What factor influence the particle density of soil?

    Particle density (ρ_p) refers to the mass of solid soil particles per unit volume, typically expressed in g/cm³. It usually ranges from 2.60 to 2.75 g/cm³ for mineral soils. Particle density is influenced by the type and composition of soil minerals rather than the pore space.

19. Mention any two soil factors affecting available water.

    Available water is the portion of soil water that plants can absorb, mainly held as capillary water. Its availability is influenced by several soil factors like soil texture, soil structure, amount of organic matter, depth of soil profile etc.

20. Name and briefly describe four common soil structure types.

    Soil structure refers to the arrangement of soil particles into aggregates or peds, which affects water movement, aeration, and root growth. Here are four common types:

    - **Granular Structure:** Small, rounded aggregates resembling breadcrumbs.
    - **Blocky Structure:** Soil forms irregular, roughly cube-shaped blocks.
    - **Platy Structure:** Description: Soil arranged in thin, horizontal plates stacked like sheets.
    - **Prismatic/Pillar Structure:** Vertical columns or pillars of soil, often capped on top.

21. What are the differences between monomineralic and polymineralic rocks.

A monomineralic rock is a rock that is composed predominantly or entirely of a single mineral. These rocks have a uniform composition, color, and texture because only one mineral type dominates.

A polymineralic rock is a rock composed of two or more different minerals. These minerals occur together in varying proportions, giving the rock a varied colour, texture, and composition.

22. Discuss how soil temperature influences overall crop growth?

Soil temperature is a critical factor affecting crop growth because it directly influences seed germination, root development, nutrient availability, and microbial activity.

PART C

Answer any one question (Each question carries 10 marks)

23. Discuss the overall significance of thermal properties of soil in agriculture.

The thermal properties of soil—including soil temperature, heat capacity, thermal conductivity, and thermal diffusivity—play a crucial role in determining crop growth, soil processes, and agricultural productivity.

1. Seed Germination

- Soil temperature affects the rate and success of seed germination.
- Each crop has an optimum temperature range; too low slows germination, too high may damage seeds.

2. Root Growth and Development

- Roots grow best in soils with suitable thermal conditions.
- Cold soils limit root activity and nutrient uptake.
- Warm soils (within optimum range) enhance root elongation and absorption.

3. Microbial Activity and Nutrient Availability

- Soil microorganisms are temperature-sensitive.
- Warm soils increase microbial decomposition of organic matter, releasing nutrients (like N, P, and K) for crops.
- Cold soils slow microbial activity, reducing nutrient availability.

4. Water Movement and Evaporation

- Soil heat influences evaporation rates, soil moisture retention, and water availability for crops.
- Hot soils may increase evaporation, leading to moisture stress.
- Thermal properties also affect capillary movement of water.

5. Soil Physical Properties

- Thermal expansion and contraction can influence soil structure, porosity, and aggregation.
- Well-structured soils allow better root penetration and aeration.

6. Crop Phenology and Yield

- Soil temperature affects plant metabolic processes, flowering, and fruiting.
- Optimal thermal conditions promote higher growth rates and yields.
- Extreme soil temperatures can stress plants and reduce productivity.

24. What is enriched compost? discuss how compost can be enriched with essential nutrients like nitrogen, phosphorus, potassium and bioinoculants?

Enriched compost is organic matter that has been decomposed and supplemented with additional nutrients or beneficial microorganisms to enhance its fertility and effectiveness as a soil amendment. Unlike ordinary compost, which primarily improves soil structure and moisture retention, enriched

compost provides essential plant nutrients and supports microbial activity, making it a more effective organic fertilizer.

Importance of Enriched Compost

- Supplies macro- and micronutrients to plants.
- Improves soil structure, aeration, and water-holding capacity.
- Enhances microbial activity, promoting nutrient cycling.
- Reduces dependence on chemical fertilizers, supporting sustainable agriculture.

Enrichment with Essential Nutrients

1. **Nitrogen (N)**
    - Nitrogen is essential for leaf growth and protein synthesis.
    - Compost can be enriched with:
        - Urea, ammonium sulfate, or farmyard manure during composting.
        - Leguminous plant residues like green manure crops (e.g., cowpea, clover).
    - These additions increase nitrogen content and promote vigorous plant growth.
2. **Phosphorus (P)**
    - Phosphorus is vital for root development, flowering, and energy transfer.
    - Enrichment methods:
        - Adding rock phosphate, bone meal, or phosphate-rich fertilizers during composting.
        - Microbial phosphate solubilizers can also enhance phosphorus availability.
3. **Potassium (K)**
    - Potassium is important for water regulation, disease resistance, and enzyme activation.
    - Enrichment methods:
        - Incorporating potash (KCl), wood ash, or crop residues rich in potassium.
        - Ensures the compost provides sufficient potassium for crop health.
4. **Bioinoculants**

- Beneficial microorganisms enhance nutrient availability and soil health.
- Examples include:
    - Rhizobium – fixes nitrogen in legume crops.
    - Azotobacter – free-living nitrogen fixer.
    - Mycorrhizal fungi – improve phosphorus uptake.
    - Phosphate-solubilizing bacteria (PSB) – make phosphorus more available to plants.
    - Adding these during composting or after maturation enriches the compost biologically.